Lab 9 - Introduction to Software Defined Radio

Analog & Digital Communication - EE322

Habib University

Dhanani School of Science & Engineering

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1 Objective

Learning to use software defined radio hardware to send and receive data using command line.

2 Overview of In-lab tasks

1. Working on A software defined radio to receive FM broadcast and Aviation broadcast.
2. Setting up command line and trying out basic command for Hack Rf one.
3. Sending data from txt file using command line, on a specific frequency.
4. Receiving data to a txt file using command line, on a specific frequency.

3 Learning outcomes

1. Familiarizing yourself with SDR hardware, understanding the basics and why we need SDR.
2. Finding practical uses of Software defined radio.
3. Exploring the posibility of digital signal processing using command line and the limited tool set. which shows the need for a sofware having more advance features for digital signal processing.

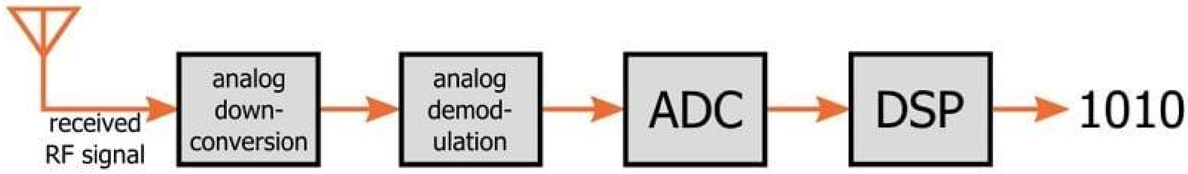
4 Introduction

- Software-defined radio is a *concept* according to which RF communication is achieved by using software (or firmware) to perform signal-processing tasks that are typically performed by hardware. *A* software-defined radio (as in, the device itself) is an RF communication system that incorporates a significant amount of this software-based signal-processing functionality.

- An SDR does not have to be a digital communication system. It may seem counterintuitive, but a very complicated digital (actually, mixed-signal) circuit board could be used to implement purely analog RF communication, such as the transmission of analog audio signals.

- An SDR does not have to provide both transmit and receive functionality. It might be only a transmitter or only a receiver. If it is capable of both transmitting and receiving, it might implement the Rx path in software and the Tx path in hardware. There is no reason why software has to be used for everything.[1]

[1] https://www.allaboutcircuits.com/technical-articles/introduction-to-software-defined-radio/



An example of a receive path in a software-defined RF data link.

5 Background

5.1 Hardware

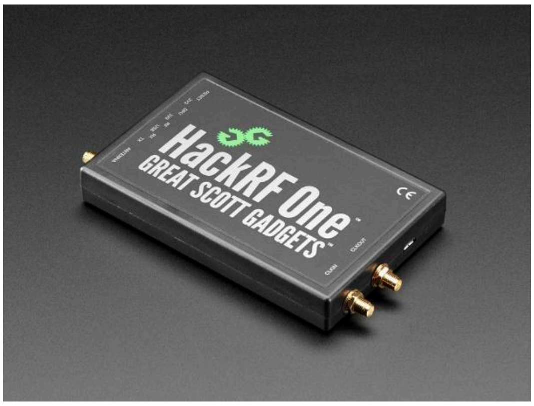


Fig: Hack rf one

HackRF One is the current hardware platform for the HackRF project. It is a Software Defined Radio peripheral capable of transmission or reception of radio signals from 1 MHz to 6 GHz. Designed to enable test and development of modern and next generation radio technologies, HackRF One is an open source hardware platform that can be used as a USB peripheral or programmed for stand-alone operation.

5.2 Features

- half-duplex transceiver
- operating freq: 1 MHz to 6 GHz
- supported sample rates: 2 Msps to 20 Msps (quadrature)
- resolution: 8 bits

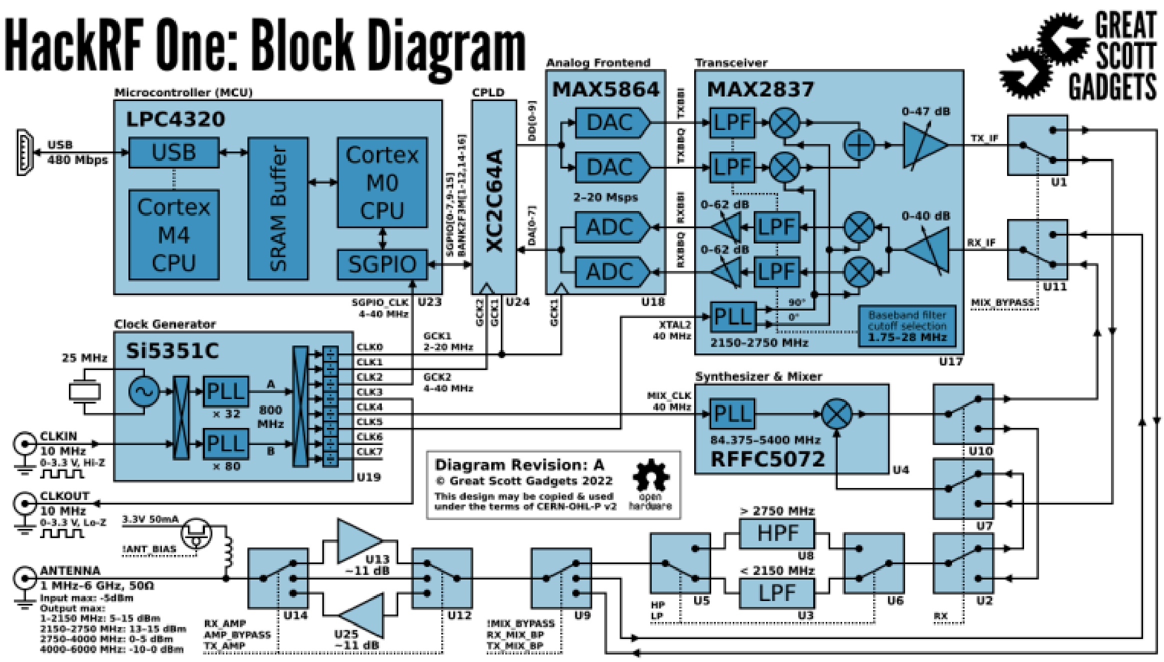


Fig: Block diagram[2]

5.3 Command line tools[3]

10.5 HackRF Tools

In addition to third party tools that support HackRF, we provide some commandline tools for interacting with HackRF. For information on how to use each tool look at the help information provided (e.g. `hackrf_transfer -h`) or the manual pages.

The first two tools (`hackrf_info` and `hackrf_transfer`) should cover most usage. The remaining tools are provided for debugging and general interest; beware, they have the potential to damage HackRF if used incorrectly.

- **hackrf_info** Read device information from HackRF such as serial number and firmware version.
- **hackrf_transfer** Send and receive signals using HackRF. Input/output can be 8bit signed quadrature files or wav files.
- **hackrf_max2837** Read and write registers in the Maxim 2837 transceiver chip. For most tx/rx purposes hackrf_transfer or other tools will take care of this for you.
- **hackrf_rffc5071** Read and write registers in the RFFC5071 mixer chip. As above, this is for curiosity or debugging only, most tools will take care of these settings automatically.
- **hackrf_si5351c** Read and write registers in the Silicon Labs Si5351C clock generator chip. This should also be unnecessary for most operation.
- **hackrf_spiflash** A tool to write new firmware to HackRF. This is mostly used for *Updating Firmware*.
- **hackrf_cpldjtag** A tool to update the CPLD on HackRF. This is needed only when *Updating Firmware* to a version prior to 2021.03.1.

6 Module: Tuning to your favorite channel of FM

For this module we will be using a software called GQRX SDR to receive data and select the appropriate demodulation to hear the Radio.

Instructions on using **GQRX SDR:**

1) When you open the software it will show a configuration page, here you will select the Hack Rf from device. you have to select the one that says "**Hack rf one**"
2) If you don't see the name in the list you may need to press the device scan button.
3) Then select the sampling rate to 8Mhz
4) Press ok.

2

3 https://hackrf.readthedocs.io/_/downloads/en/latest/pdf/

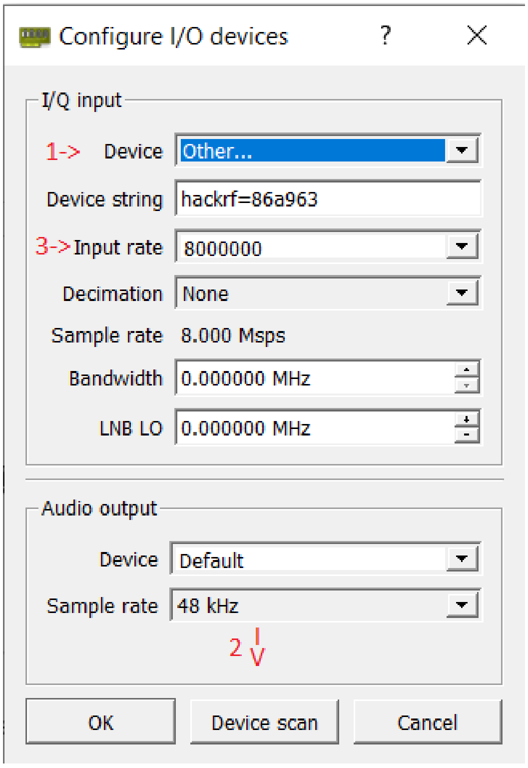


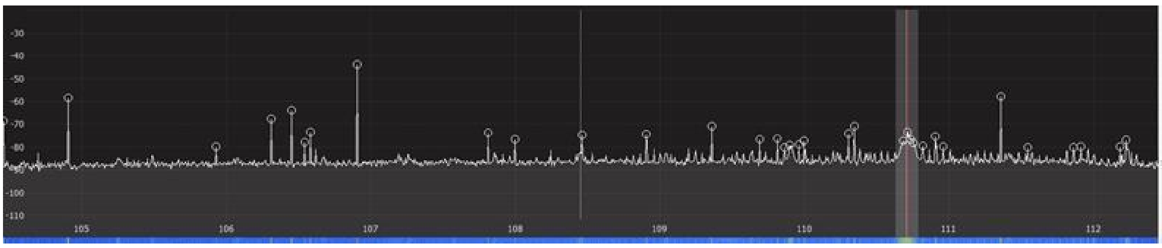
Fig: startup screen

5) By selecting the play button, we can toggle on or off receiving. If we require to go back to the hardware setup we can select the second option next to the play button.

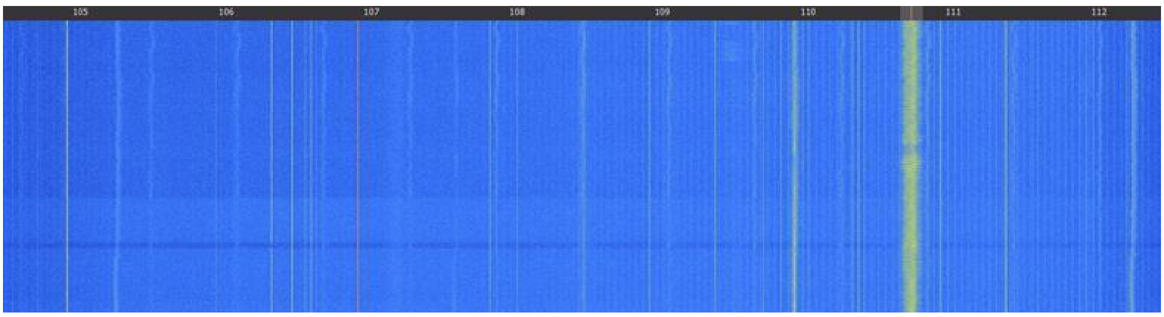
Fig: player options

Part II

1) Graph one is Power (dB) Frequency graph(kHz)



2) Graph two is called waterfall frequency graph with time on the y axis and frequency on the x axis power is displayed by color.



3) The numbers that a present above the graph (for frequency of the received signal) can be edited by click on one of the number and then using your arrow keys to increase (up key), decrease (down key), move left(left key) and move right (right key). Also you can enter a number.

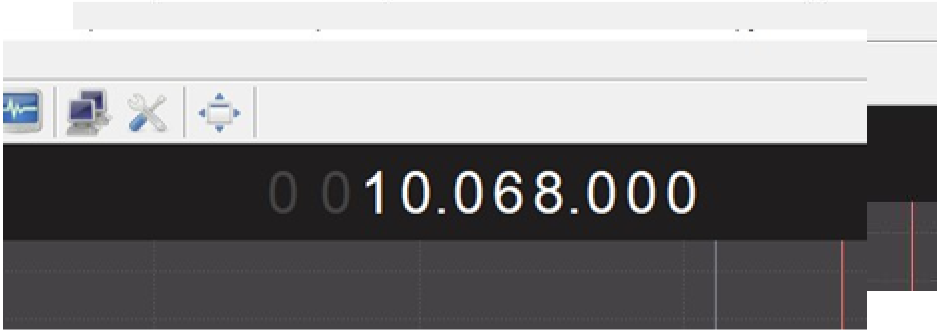


Fig: frequency dial

4) This is the receiver options. We will focus on the mode option. This will be the demodulation setting. For FM band we will keep the option to be WFM (stereo).

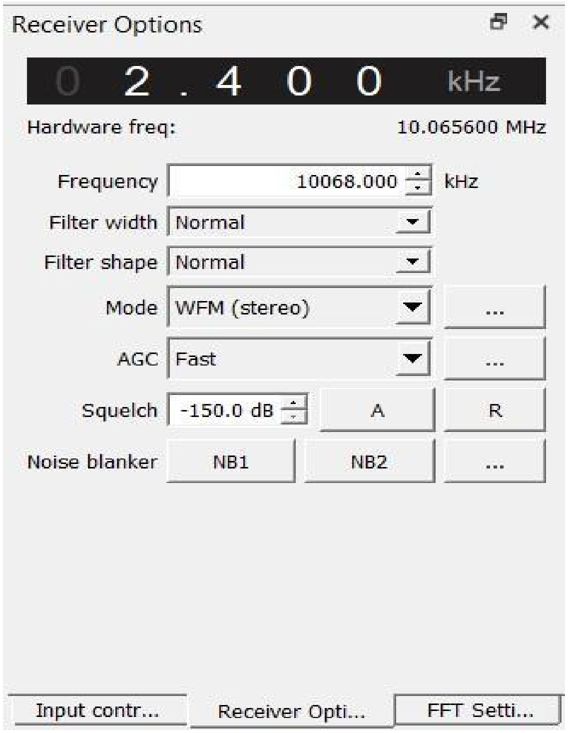


Fig: receiver options

5) To adjust the filter required for the particular band by scrolling the cursor with the horizontal transparent box. Dragging the box increases the size, which would allow more frequencies to pass. This may make the signal noisier.

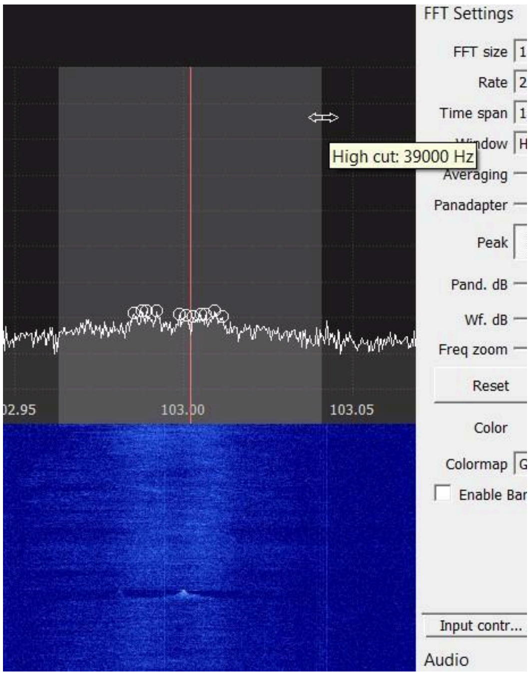


Fig: Changing filter length

6) In FFT settings we will enable peak detect to make it easier to detect signals.

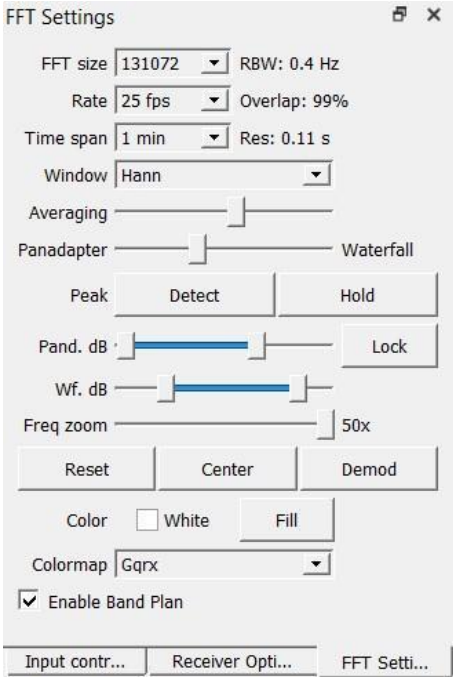


Fig: FFT options

7) Sometimes the signal is too weak to be seen on the spectrum. We can increase e the gain of the system by adjusting the IF and BB amplifiers.

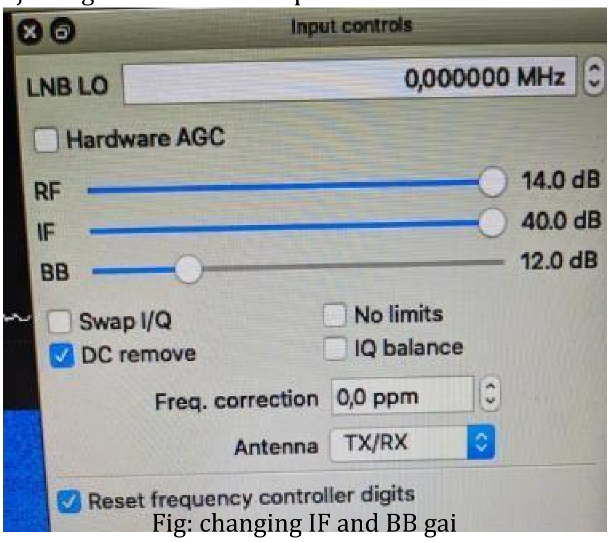
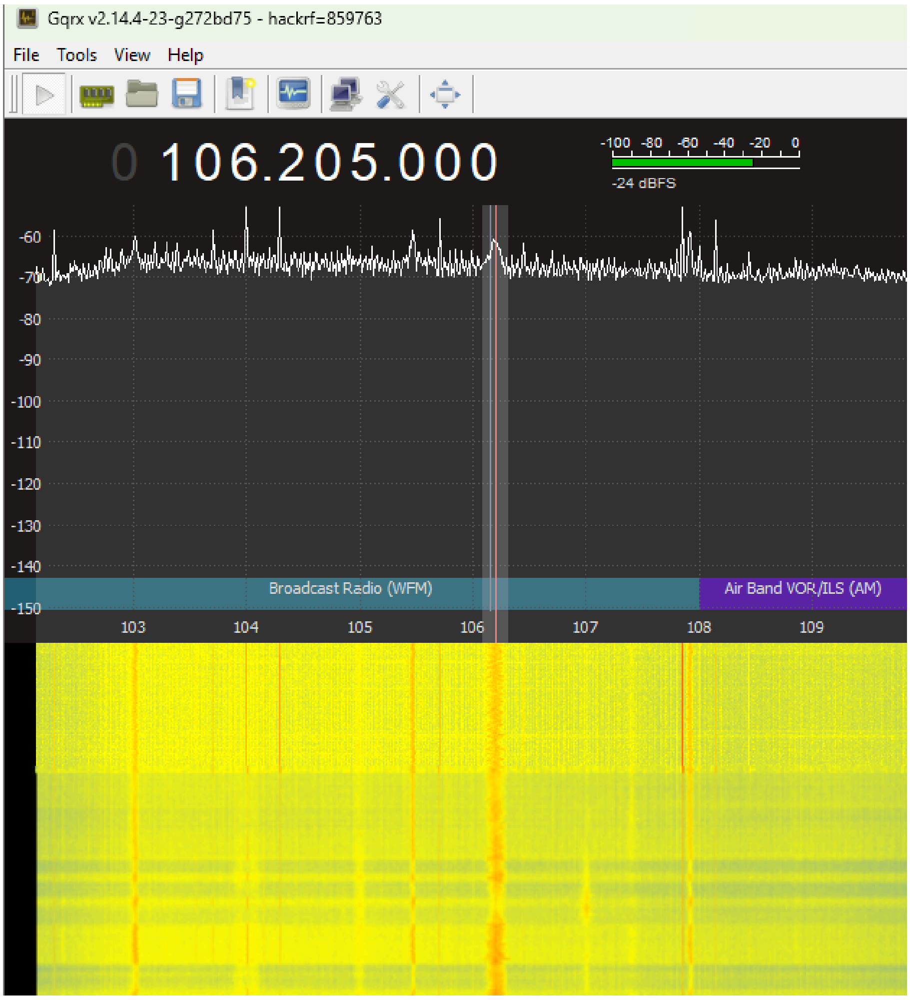


Fig: changing IF and BB gai

6.3 Task 1: Find a FM broadcasting frequency between 90 Mhz to 108 Mhz. Attach a snippet.



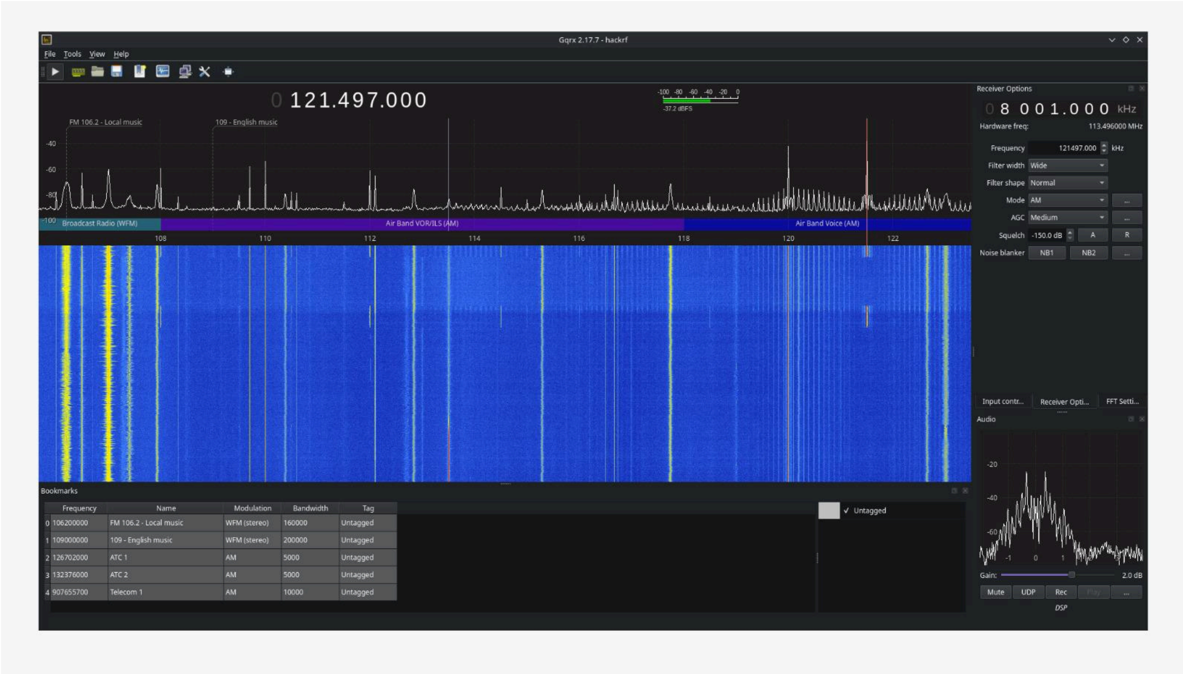
Task 2: Find the broad casting frequency for Aviation band.

Approach (APP): 125.500 MHz
ATIS: 126.700 MHz
Ground (GND): 121.600 MHz
Tower (TWR): 118.300 MHz

Task 3: Find modulation type for Aviation band.

Amplitude Modulation

Task 4: Using your answers form previous questions design a way to receive radio chatter of the specific aviation TWR frequency. Attach a snippet.



More about antennas

There are two types of antennas **dipole antenna (**λ/2**)** and **monopole antenna (**λ/4**).** Monopole antennas are more commonly used. Some types that you may be familiar are.

Whip antenna

Rubber Ducky antenna

Telescopic antenna

Task 5: Finding the length of the antenna required for the FM frequency that you tuned to.

Let's figure out the wavelength for **7.150** MHz since we are going to divide that into two for a half-wavelength.

The speed of light is 984,000,000 feet per second.

- Divide 984,000,000 feet per second by 1,000,000 = 984 for one wavelength calculation in MHz.
    - Divide 984 by 4 = 246 for a Quater-wavelength in MHz
- Multiply 246 by .9512 velocity of propagation or end effect in copper conductors = 234
    - Divide 234 by **7.150** MHz = 33 feet

Now calculate the antenna length for your frequency:

- Speed of Light = 984,000,000 ft/s
- 984,000,000/1000000 = 984 for one wavelength calculation in MHz
- 984/4 = 246 for a quarter wavelength in MHz
- 246 times 0.9512 velocity of propagation in copper conductors = 234
- 234/121.497 MHz = **1.92 feet**

7 Module: Exploring hardware with command line

7.3 Task 1: Test if Hack Rf is connected with your device.

Steps:

1) Open Command prompt
2) Enter “**hackrf_info**”

```
C:\Users\DELL>hackrf_info
hackrf_info version: git-7047252
libhackrf version: git-7047252 (0.6)
Found HackRF
Index: 0
Serial number: 000000000000000057b068dc2386a963
Board ID Number: 2 (HackRF One)
Firmware Version: 2018.01.1 (API:1.02)
Part ID Number: 0xa000cb3c 0x00464366
```

Fig: Information about the device

Once you have received similar results shown in the image above. You will set up some files to receive and transmit data. Following commands are used for setting up the files using command line.

Note: You need to make sure that you are in the correct directory.

```
C:\Users\DELL>cd Downloads

C:\Users\DELL\Downloads>
```

Fig: change directory

```
C:\Users\DELL\Downloads>mkdir communication

C:\Users\DELL\Downloads>cd communication
```

Fig: making directory

```
C:\Users\DELL\Downloads\communication>echo HelloWorld > transmit.txt
```

Fig: Making transmit file with “HelloWorld” as text

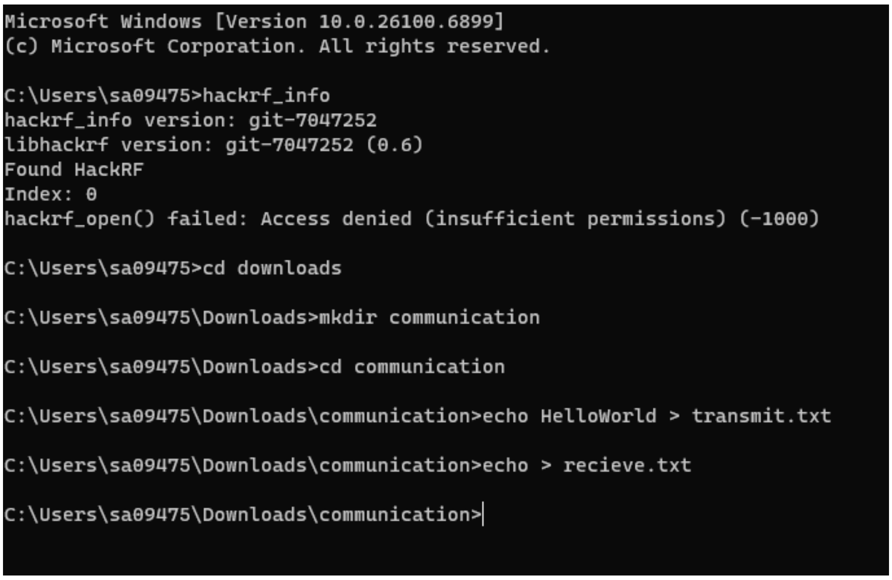
```
C:\Users\DELL\Downloads\communication>echo > recieve.txt
```

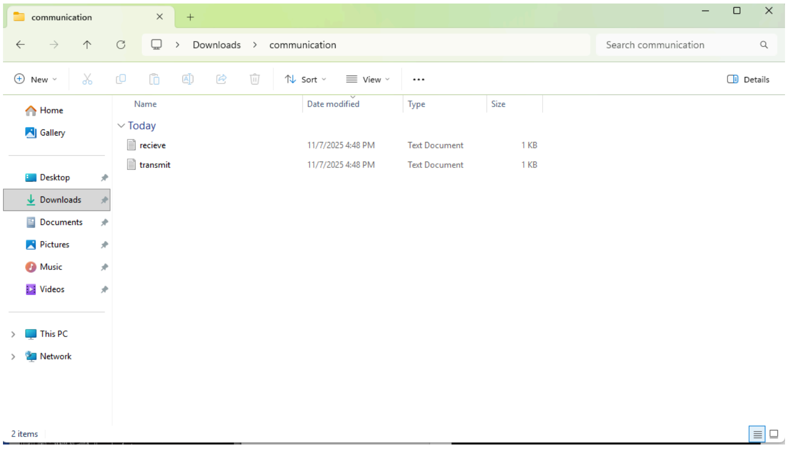
Fig making receive txt file

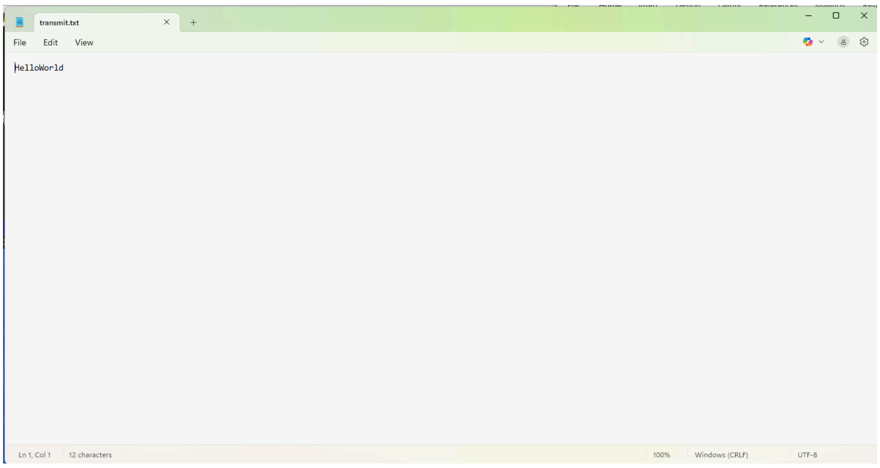
Q) Add snippets of hackrf_info and the directory with the two files.

- ☐

Below are the snippets attached.


Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.
C:\Users\sa09475>hackrf_info
hackrf_info version: git-7047252
libhackrf version: git-7047252 (0.6)
Found HackRF
Index: 0
hackrf_open() failed: Access denied (insufficient permissions) (-1000)
C:\Users\sa09475>cd downloads
C:\Users\sa09475\Downloads>mkdir communication
C:\Users\sa09475\Downloads>cd communication
C:\Users\sa09475\Downloads\communication>echo HelloWorld > transmit.txt
C:\Users\sa09475\Downloads\communication>echo > recieve.txt
C:\Users\sa09475\Downloads\communication>


communication
Downloads > communication
Search communication
New
Sort
View
Details
Home
Gallery
Desktop
Downloads
Documents
Pictures
Music
Videos
This PC
Network
Name
Date modified
Type
Size
Today
recieve
11/7/2025 4:48 PM
Text Document
1 KB
transmit
11/7/2025 4:48 PM
Text Document
1 KB
2 items


transmit.txt
File
Edit
View
HelloWorld
Ln 1, Col 1
12 characters
100%
Windows (CRLF)
UTF-8

Q) Which devices can be listened to by using Hack RF? Can we transmit and receive signals simultaneously?

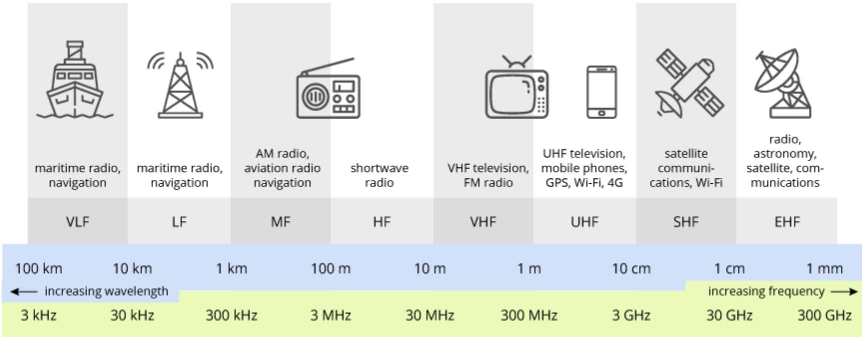


Fig: radio frequency spectrum

HackRF One can receive and transmit signals in roughly the HF to lower SHF range (1 MHz – 6 GHz). That includes:

- Shortwave / HF radio

- VHF: FM radio, TV, aircraft, maritime

- UHF: mobile, GPS, Wi-Fi, Bluetooth

- Lower SHF: 5 GHz Wi-Fi, ISM bands

Cannot receive: VLF, LF, most MF, and EHF bands.

Q) What is the importance of an SDR and what applications do you think an SDR can be used for?

A Software Defined Radio (SDR) is important because it makes radios flexible, cost-effective, and easy to upgrade—hardware functions are done in software, so one device can handle many frequencies and protocols. It can be used for:

- Amateur (ham) radio

- Wireless research (4G/5G, IoT)

- Signal monitoring and intelligence

- Aircraft/ship tracking (ADS-B, AIS)

- FM/TV/satellite reception

- Emergency and disaster communications

It's basically a universal, adaptable radio in software form.

Name & ID:____________________________ **Date:** ____________

EE-322L Analog and Digital Communications Lab
Fall 2025
Habib University
Dhanani School of Science & Engineering

LAB 09: SDR

Lab #09 Marks distribution:

		LR1=20	LR4=30	LR5=30	AR4=20
In-Lab Tasks	**Module 6**	/10	/15	/15	/20
	Module 7	/10	/15	/15	
Marks Obt.	**/100**				

Lab Evaluation Assessment Rubric

#	Assessment Elements	(0<Level 1<=4)	(4< Level 2<=6)	(6< Level 3<=8)	(8< Level 4<=10)
LR1	**Neat and Clean Circuit layout**	Components are wired and didn't show neat and clean connections and minimal efforts shown.	Components are wired with untidy connection and didn't show neat and clean connections	Few but not all Components are wired with neat and clean connections	Complete components are wired with neat and clean tight connections. Given task completed in due time.
LR3	**Troubleshooting**	Unable to identify the fault/minimal effort shown	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and able to make necessary steps and actions to correct it.
LR4	**Data Collection**	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included	Measurements are somewhat inaccurate and very imprecise. Observations are incomplete or recorded in a confusing way. Major errors using symbols, units and significant digits	Measurements are mostly accurate. Observations are generally complete. Minor errors using symbols, units and significant digits	Measurements are both accurate and precise. Observations are very thorough. Includes appropriate symbols, units and significant digits and completed in due time.
LR5	**Results & Plots**	Figures, graphs, tables contain errors or are	Most figures, graphs, tables OK, some still missing	All figures, graphs, tables are correctly drawn,	All figures, graphs, tables are correctly drawn and contain titles/captions.

		poorly constructed, have missing titles, captions, units missing or incorrect, etc.	some important or required features	but some have minor problems or could still be improved	
LR8	**Contribution**	No participation towards the group tasks	Slight participation towards the group tasks	Substantial participation towards the group tasks	Outstanding participation towards the group tasks
AR4	***Report Submission**	Late submission after 1 week.	Late submission after 4 days and within a week.	Late submission after the lab timing and after 2 days of the due date.	Timely submission of the report and in lab time.

***Report:** Report will not be accepted after 1 week of due date